Reminders and announcements

- During lectures you can submit questions to: vevox.app ID: 107-018-952
- You can find answers to questions in the COMP1870 Teams Channel (link in Minerva).
- Note that the deadline for portfolio submissions is 2.00pm, 10 days after the worksheet is posted.
  - We expect and encourage everyone to make their portfolio submissions by the deadline.
  - There is another submission portal you can use for late submissions and resubmissions.

Revision

Name	**Previous statement(s)**	**New statement**	**Justification**
Conjunction Elimination	$j.\ A \wedge B$	A	$\wedge$Elim j
	$j.\ A \wedge B$	B	$\wedge$Elim j
Conjunction Introduction	$j.\ A$ $k.\ B$	$A \wedge B$	$\wedge$Intro j, k
Implication Elimination	$j.\ A \Rightarrow B$ $k.\ A$	B	$\Rightarrow$Elim j, k
Implication Introduction	$\vert\ _\ j.A$ $\vert\ \ \vdots$ $\vert\ \ k.B$	$A \Rightarrow B$	$\Rightarrow$Intro j-k
Disjunction Introduction	$j.\ A$	$A \vee B$	$\vee$Intro j
	$j.\ A$	$B \vee A$	$\vee$Intro j
Disjunction Elimination	$j.\ A \vee B$ $k.\ A \Rightarrow C$ $l.\ B \Rightarrow C$	C	$\vee$Elim j, k, l
Negation Introduction	$j.\ A \Rightarrow (B \wedge (\neg B))$	$\neg A$	$\neg$Intro j
Double Negation	$j.\ \neg(\neg A)$	A	$\neg\neg\ j$
	$j.\ A$	$\neg(\neg A)$	$\neg\neg\ j$

Proofs

Prove $(A \Rightarrow B) \therefore (\neg B) \Rightarrow (\neg A)$

Mathematics is characterised by the use of **deductive logic** to **prove** statements. Constructing proofs is an art as much as a science, and is a difficult skill to learn.

There are many statements that mathematicians do not know how to prove:

- **Riemann hypothesis.** Every non-trivial zero of the Riemann zeta function has real part $\frac{1}{2}$.
- **P $\neq$ NP.** There are problems that can be solved in polynomial time by non-deterministic Turing machines but not by deterministic Turing machines.
- **Goldbach's conjecture.** Every even number greater than two can be expressed as the sum of two primes.

Three important proof styles

- **Direct proof.** We want to prove a statement of the form $X \Rightarrow Y$.

  We assume that X is true and then deduce a sequence of statements, ultimately arriving at Y.

- **Contrapositive proof.** We want to prove a statement of the form $X \Rightarrow Y$.

  We give a direct proof of the equivalent statement $(\neg Y) \Rightarrow (\neg X)$.

- **Proof by contradiction.** We want to prove a statement X.

  We prove that $\neg X$ implies something impossible. That is, we give a direct proof of a statement with the structure $(\neg X) \Rightarrow (P \wedge (\neg P))$. This means $\neg(\neg(X))$ is true.

Example of direct proof

The **integers** are the numbers $\ldots, -2, -1, 0, 1, 2, \ldots$.

Proposition 3.3. Let a and b be integers. If a and b are even, then $a + b$ is even.

Proof. We assume a and b are both even. This means $a = 2m$ and $b = 2n$ for some integers m and n.

Now $a+b = 2m + 2n = 2(m+n)$.

Because m and n are integers, so is $m+n$.

Since $a+b$ can be expressed as the product of 2 and an integer, $a+b$ is even. □

Example of contrapositive proof

Proposition 3.4. Let z be an integer greater than one. If $2^z - 1$ is prime, then z is prime.

Proof. $X: 2^z - 1$ is prime

$Y: z$ is prime

We want is $X \Rightarrow Y$. We try to prove $(\neg Y) \Rightarrow (\neg X)$.

We assume $\neg Y: z$ is not prime.

Because z is greater than 1 and non-prime it can be factored $z = kx$ where $1 < k, x < z$.

Example of contrapositive proof

Now $(2^x-1)(1+2^x+2^{2x}+\cdots+2^{(k-1)x})$

$= (2^x+2^{2x}+2^{3x}+\cdots+2^{kx})$

$\quad -(1+2^x+2^{2x}+\cdots+2^{(k-1)x})$

$= 2^{kx}-1 = 2^z-1$

So 2^z-1 is divided by 2^x-1.

Because $1<x<z$, we have $1<2^x-1<2^z-1$.

Therefore 2^z-1 is non-prime. $\square$

Example of proof by contradiction

Theorem 3.5. There are infinitely many prime numbers.

Proof by contradiction. We assume this is false. Therefore there is a finite list of primes: $p_1, p_2, p_3, \ldots, p_n$.

Let z be $p_1 p_2 p_3 \cdots p_n + 1$.

Because z is an integer greater than 1, it is divisible by a prime (we will prove this soon). This prime is on our list, so we assume p_i divides z.

So $\frac{z}{p_i}$ is an integer.

Example of proof by contradiction

Theorem 3.5. There are infinitely many prime numbers.

$$\frac{z}{p_i} = \frac{p_1 p_2 p_3 \cdots p_n + 1}{p_i}$$

$$= \frac{p_1 p_2 \cdots \cancel{p_i} \cdots p_{n-1} p_n}{\cancel{p_i}} + \frac{1}{p_i}$$

$$= \underbrace{p_1 \cdots p_{i-1} p_{i+1} \cdots p_n}_{\text{integer}} + \underbrace{\frac{1}{p_i}}_{\text{non-integer}}$$

So $\frac{z}{p_i}$ is not an integer. This contradicts the conclusion that it is an integer. We reject the assumption there a finite primes $\square$

Inductive proofs

The **positive integers** are the numbers $1, 2, 3, \ldots$. The **non-negative integers** are $0, 1, 2, \ldots$.

- We typically use a **proof by induction** to prove that some statement is true for all positive integers, or for all non-negative integers.
- We start by proving that the property holds for 1 (in the positive integer case) or 0 (in the non-negative integer case). This step is called **checking the base case**.
- We then prove that if the statement is true up to an arbitrary point in the integers, then it is true for the next integer: if it is true for the numbers $1, 2, \ldots, k$, then it is true for $k + 1$, where k is an arbitrary integer.
- If we can accomplish these steps, we have proved the statement is true for all positive (non-negative) integers.

Inductive proofs

An equivalent explanation, using predicate logic.

- We want to prove that $\forall n\ P(n)$ is true, where the domain is the set of positive integers and P is some predicate.
- We check $P(1)$ is true.
- We prove that when k is an arbitrary positive integer,

$$P(1) \wedge P(2) \wedge P(3) \wedge \cdots \wedge P(k) \Rightarrow P(k+1)$$

- Now $P(n)$ is true for all positive integers n.

Inductive assumption

How do we prove

$$P(1) \wedge P(2) \wedge P(3) \wedge \cdots \wedge P(k) \Rightarrow P(k+1) \ ?$$

We assume $P(1) \wedge P(2) \wedge P(3) \wedge \cdots \wedge P(k)$ is true, and use this assumption to derive $P(k+1)$.

In other words, we assume that the statement is true for all the integers $1, 2, \ldots, k$. This is called the **inductive assumption** or **inductive hypothesis**.

We use this assumption to prove that the statement is true for $k+1$.

Example of an inductive proof

Proposition 4.1. Let n be a positive integer. Then

$$1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}$$

Proof. We check the base case $n = 1$. The left side is equal to 1. The right side $\frac{1(1+1)}{2} = 1$. The base case is true.

We make the inductive assumption that the equation is correct when n is in the range $1, \ldots, k$ for an arbitrary positive integer k.

Example of an inductive proof

We use this to prove the equation is correct when $n = k+1$.

When $n = k+1$, the left side is

$$1 + 2 + \cdots + k + (k+1)$$

The inductive assumption says $1+2+\cdots+k = \frac{k(k+1)}{2}$. We substitute this in $1+2+\cdots+k+(k+1)$

$$= \frac{k(k+1)}{2} + (k+1) = \frac{k(k+1)}{2} + \frac{2(k+1)}{2}$$

$$= \frac{k(k+1)+2(k+1)}{2} = \frac{(k+1)(k+2)}{2} = \frac{(k+1)((k+1)+1)}{2}$$

This is the expression we get if we substitute $n = k+1$ into $\frac{n(n+1)}{2}$. So the equation is also true when $n = k+1$. □

end of proof

Prime decomposition

Here is an inductive proof using the idea of a **minimal counterexample**.

Theorem 4.4. Any integer greater than 1 is either prime or is equal to a product of prime numbers.

Proof. We assume this is false. Therefore there is a counterexample (an example that makes this statement false). Let x be the smallest counterexample. So x is an integer greater than 1, x is not prime or a product of primes.

Prime decomposition

Since x is greater than one and not prime, it can be factored $x = yz$, where $1 < y, z < x$

(Both y and z are greater than 1 and less than x.)

Since $y, z < x$ they are not counterexamples! So we can write:

Prime decomposition

$y = p_1 p_2 \cdots p_n$, $z = q_1 q_2 \cdots q_m$, where $p_1, \ldots, p_n, q_1 \ldots q_m$ are prime.

But $x = yz = p_1 \cdots p_n q_1 \cdots q_m$.

So x is a product of primes.

This contradicts the assumption that x is a counterexample. ∎

Inductive proofs in computer science

We frequently use inductive proofs in computer science in these applications:

- analysing running time of algorithms, and
- proving the correctness of algorithms.

Analysing running times

Imagine that an algorithm has running time $T(n)$ when the input size is n.

We know that $T(1) = 1$ and $T(2) = 2$, and $T(n) = T(n-1) + 2T(n-2)$ when $n > 2$.

n	$T(n)$
1	$1 = 2^0$
2	$2 = 2^1$
3	$T(2) + 2T(1) = 2 + 2 \cdot 1 = 4$
4	$T(3) + 2T(2) = 4 + 2 \cdot 2 = 8$
5	16

Analysing running times

Theorem: $T(n) = 2^{n-1}$.

Proof: by induction.

Base case:

Analysing running times

Inductive step:

Analysing running times

Weak/strong induction

Sometimes we do not need the full inductive assumption.

If we only use the assumption that $P(k)$ is true, rather than the full assumption that $P_1, P_2, \ldots, P(k)$ are all true, we have used **weak induction**.

Otherwise we have used **strong induction**.